

Shaantanu Iyengar

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Education

University of Bath

Expected June 2028

MEng Electronic Engineering with Space Science and Technology

- Relevant Modules: Circuit Theory, Digital Electronics, Control Systems, Engineering Design, Programming

Experience

Systems Engineering Intern - Leonardo, UK

Jun - Aug 2026

- Operating as a systems engineer within a cross-disciplinary team, generating a synthetic RF dataset across a continuous 6-stage RF-to-storage pipeline
- Designed the signal-chain models in Simulink and deployed it to an RFSoc, optimising execution via compiled Accelerator-mode simulation to cut per-scenario runtime by **70%**
- Engineered a MATLAB App (MLApp) serving as the primary user interface for the full RF simulation pipeline, exposing 12 configurable impairment parameters that drive the downstream models and hardware in real time
- Developed the VHDL data capture logic for an FPGA, implementing a double-buffered (ping-pong) architecture to eliminate capture stalls during transfer via the VisualAnalog API, increasing capture throughput by **40%** and reducing capture-to-storage latency by **50%**
- Conducted functional analysis in MATLAB System Composer to derive **25+** system requirements from 16 stakeholder needs, then built a VCRM covering 30 test cases, achieving **98%** requirement verification

Projects

ML-Based Component Health monitoring - Space System

Feb 2026

- Designed an object-oriented Python system for ML-based component health prognostics, targeting predictive failure detection in space systems with no post-launch maintenance access, validated on NASA's C-MAPSS turbofan degradation dataset as a proof of concept
- Built a triple-mode ML pipeline predicting component failure timelines and detecting degradation patterns across sensor data, then tuned Random Forest hyperparameters via grid search, cutting prediction error by **15%** and increasing accuracy by **11%**

180-Degree Ultrasonic Radar System

Nov 2025

- Engineered a 180-degree radar system using an Arduino in conjunction with an ultrasonic sensor to display distances to targets on an LCD screen, developing proficiency in hardware-software interfacing
- Calibrated the sensor offset to improve measurement precision from ± 10 cm to ± 6 cm, addressing errors caused by signal reflections and environmental factors

Microprocessor System Design - FPGA Integration Project

Mar 2026

- Integrated a custom microprocessor system on an FPGA using SystemVerilog, designing and combining key processor components, including an Arithmetic Logic Unit (ALU), PC and registers
- Implemented and tested an assembly-level instruction set, verifying correct operations through switch and LED interaction, resulting in successful execution by producing visual output through a VGA display

Synchronous Buck Converter - Circuit and PCB Design Project

Apr 2026

- Designed a synchronous buck converter in KiCad, iterating from an initial 2-layer prototype to a revised 4-layer board, applying integrity analysis to cut down thermal resistance by over **50%** and shrinking a critical voltage spike by over **94%**
- Resolved a shoot-through risk through routing optimisation, raising the switching-noise frequency by **170%** without altering the circuit's components or topology

Skills

Programming languages: MATLAB, Python, C++, VHDL, SystemVerilog

Technical Skills and Tools: FPGA Design (Vivado, Quartus Prime), PCB & Circuit Design (KiCAD, OrCAD), Simulink, System Composer, Autodesk Inventor, Arduino prototyping, Azure DevOps